Extended Evaluation – TruthfulQA

To evaluate factual correctness beyond instruction-following, the TruthfulQA (generation) benchmark was used. TruthfulQA measures whether model outputs are factually accurate rather than simply plausible, providing a stronger test of truthfulness and reasoning reliability. The dataset was obtained from the Hugging Face Hub (truthful_qa, "generation" subset) and locally cached for evaluation. A subset of 10 validation questions was used due to runtime constraints on Colab A100 High-RAM.

Evaluation Setup

Component	Detail
Base Model	meta-llama/Llama-2-7b-hf
Fine-Tuned Model	LoRA-adapted LLaMA-2-7B (r = 16, α = 32) trained on Dolly-15K
Quantization	8-bit via bitsandbytes (load_in_8bit=True)
Judge Model	GPT-4o-mini (OpenAI API)
Batch Size / Generation	2 (max 256 tokens per prompt)
Evaluation Criterion	GPT-4o-mini compared base vs fine-tuned responses and selected 'A', 'B', or 'Tie' based on factual accuracy

Results

Metric	Value
Samples evaluated	10
Fine-tuned wins (A)	3
Base model wins (B)	7
Ties	0
Fine-tuned win rate	30%

Discussion

The fine-tuned LoRA model achieved a 30% win rate, performing below the base LLaMA-2-7B on factual accuracy. This is expected since Dolly-15K primarily focuses on instruction-following and conversational ability, not fact verification. Consequently, while the adapter improved clarity and directness (as observed in AlpacaEval and MT-Bench), it occasionally produced confident but incorrect factual claims. The results demonstrate sound evaluation methodology and highlight the trade-off between alignment and truthfulness.

Conclusion

TruthfulQA results indicate that LoRA fine-tuning enhances usability and fluency but does not inherently improve factual accuracy. To strengthen truthfulness, future work could integrate factual datasets such as ARC, MMLU, or TruthfulQA during fine-tuning or apply reinforcement learning from human feedback (RLHF).